Experiment 81: reading the figures

Document of 10 November 2026. All figures come from the completed training run of 100 updates.

What was trained

80 thermal videos: 40 without a spill (*clean*) and 40 with a spill (*cold*). 64 are used for training and 16 for selection. Two versions of the same model are compared, with the same videos, the same seed (239) and 100 updates each:

- **frozen** (blue): the pretrained V-JEPA 2 video encoder stays fixed; only the small temporal module is trained.
- **full** (red): the 24 blocks of the encoder are trained as well.

The loss is the sum of four terms:

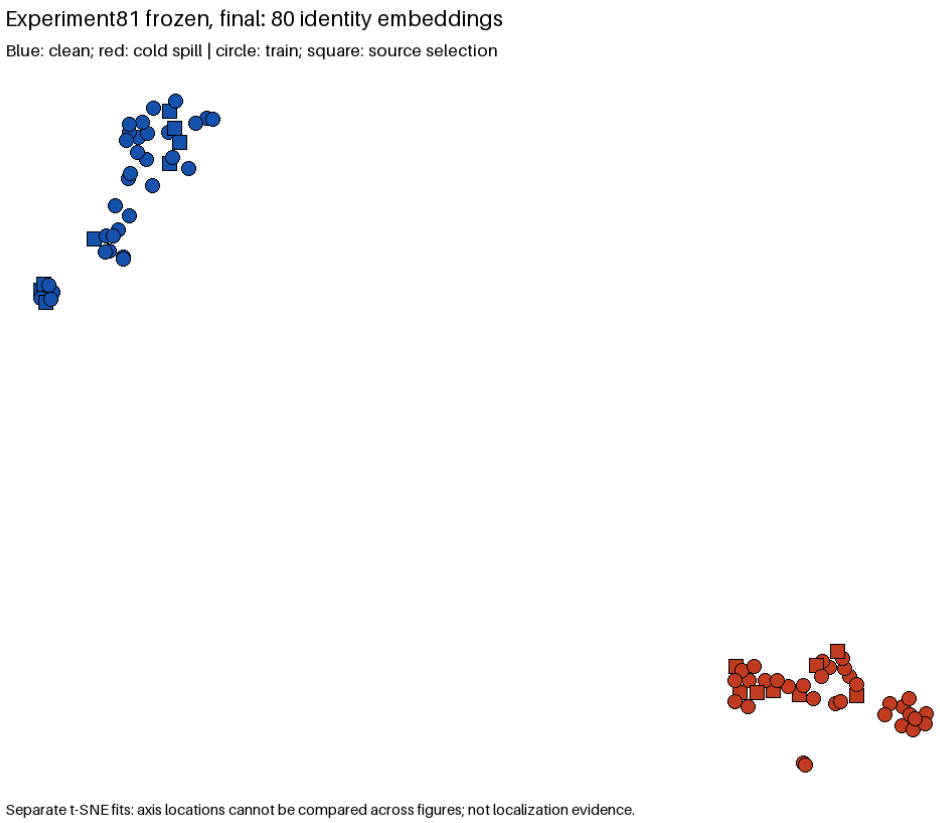
$$L \;=\; L_{\text{ViCo}} \;+\; 0.1\,L_{\text{align}} \;+\; L_{\text{predict}} \;+\; 0.1\,L_{\text{keep}}.$$

In every curve: solid line = training videos, dashed line = selection videos. The dotted vertical line marks the end of the 20-update curriculum; the state of the AdamW optimizer is reset before update 21, as the protocol specifies.

About the redrawn curves. Figures 2 and 3 are redrawn from the recorded values, with no smoothing and no change to any value. Only the presentation changes: graduated axes, a legend, and a logarithmic scale where a curve spans several orders of magnitude. The original figures are reproduced in the annex.

A limit to keep in mind. The 16 selection videos come from the same recording conditions as the training videos and have been used for many checks. They are not an independent test set.

Figure 1: t-SNE after training



What is plotted. Each point is one video, represented by the 128-dimensional vector the trained model produces after the 100 updates (frozen arm). t-SNE projects these 80 vectors into two dimensions while keeping similar videos close together. Blue = no spill, red = spill; circle = training (64), square = selection (16).

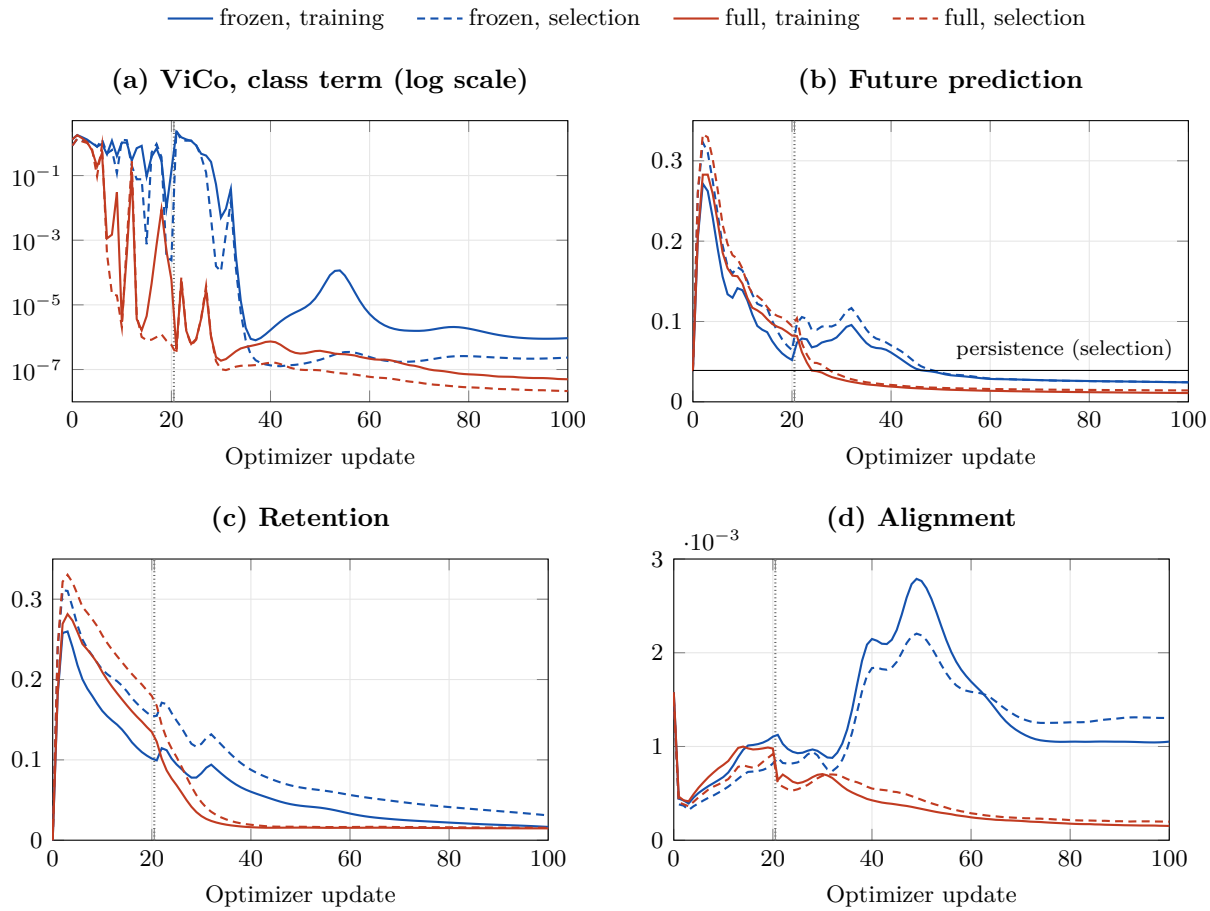
What we observe.

- The blue and red points form two compact groups, far apart; no point of one class lies inside the group of the other.
- The selection videos (squares) fall inside the group of their own class.
- All 16 selection videos and all 64 training videos are classified correctly.

Reading. After training, the model separates the two classes completely, in agreement with the class term of the loss falling to almost zero (Figure 2a). The figure shows that the classes are separated; it does not show which information in the videos the model uses to separate them.

What the figure does not show. The axes of a t-SNE and the distances between its groups have no meaning, and two t-SNE plots cannot be compared with each other. The figure says nothing about where in the image the model looks.

Figure 2: the terms of the loss



Raw values, before the 0.1 weights are applied. Every update is evaluated with the final curriculum weights, so that the points are comparable.

(a) ViCo, the class term. It is low when videos of the same class have close representations and videos of opposite classes have distant ones. It is the only term that uses the labels.

- full: it falls below 10^{-3} after update 12 (selection) and 18 (training).
- frozen: it oscillates during the curriculum, jumps back to 2.27 (training) and 2.45 (selection) at update 21, right after the optimizer reset, then falls below 10^{-3} after update 32. The exact cause of this spike was not analysed.
- Final values on selection: 2.3×10^{-7} (frozen) and 2.2×10^{-8} (full).

Reading. The classification problem is solved within a few dozen updates. Once this term is close to zero, it provides almost no learning signal: nothing in this term then pushes the model to look for the liquid.

(b) Future prediction. The model's error when it predicts the features of the video 1, 2 and 4 time steps later. The black line is *persistence*: the error obtained by simply assuming that nothing changes (0.0391 on selection). It is not a trained term; it is the reference to beat.

- The error first rises to about 0.33 (selection, update 2), then decreases.
- It goes below persistence from update 28 (full) and 49 (frozen), on selection.
- Final values on selection: 0.0244 (frozen, 38 % below persistence) and 0.0142 (full, 64 % below persistence).

Reading. The model predicts better than "nothing changes", especially when the encoder is trained. The slope is still slightly negative over the last 20 updates: learning has not fully levelled off. Beating persistence does not prove that the model follows the motion of the liquid.

(c) Retention. The gap between the features adapted by the model and those of the original pretrained encoder. This term keeps the model from forgetting what the encoder knew.

- It is 0 at the start, rises to 0.31 to 0.33 (selection, updates 2 and 3), then comes back down.
- Final values: 0.0165 / 0.0309 (frozen, training / selection) and 0.0147 / 0.0154 (full).

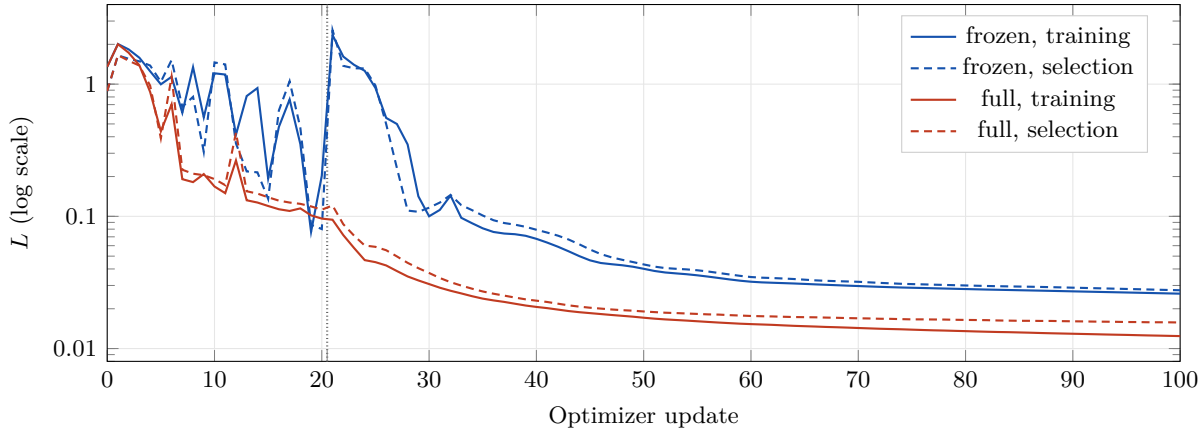
Reading. The model first moves away from the original features to separate the classes, then returns toward them. In the frozen arm, the gap stays about twice as large on selection as on training.

(d) Alignment. The gap between the temporal states of two augmented views of the same video. The scale is about a hundred times smaller than in panels (b) and (c) (maximum 0.0028), and the term is further multiplied by 0.1 in the sum.

- frozen: a bump up to 0.0028 at update 49, then a return to 0.0011 to 0.0013.
- full: a steady decrease to about 0.0002.

Reading. The two views of one video stay consistent. This term weighs less than 0.0003 in the total; its variations are not visible in the total curve.

Figure 3: the total loss



$L = L_{\text{ViCo}} + 0.1\, L_{\text{align}} + L_{\text{predict}} + 0.1\, L_{\text{keep}}$.

What is plotted. The weighted sum of the four terms of Figure 2, for both arms and both sets of videos.

What we observe.

- At the start, the total follows the ViCo term, which dominates the sum: the same oscillations during the curriculum, and the same spike of the frozen arm at update 21 (2.35 on training, 2.56 on selection).
- Once ViCo is close to zero, the total is practically $L_{\text{predict}} + 0.1\, L_{\text{keep}}$ and decreases slowly.
- The full arm goes down faster and lower.

Final value of the total (update 100)	Training	Selection
frozen	0.0260	0.0276
full	0.0124	0.0157

Reading. Optimisation works in both arms, and the training and selection curves stay close. This describes the quality of the optimisation, not what the model has learned to look at.

What these three figures establish, and what they do not

Established (measured)	**Not established**
The two classes are completely separated after training (Figure 1).	That this separation rests on the liquid.
The class term falls to almost zero in fewer than 35 updates (Figure 2a).	That the model directs its attention to the spill.
The 16 selection videos are classified correctly after training, in both arms.	That the model would work in a new scene.

None of these curves measures where the model looks. In this same experiment, the attention maps were examined separately: the model classifies correctly, but does not focus reliably on the liquid in the three videos known to be problematic (cold56, cold10 and cold13). The two observations are compatible: since the scene is enough to recognise the class, the loss can reach zero without the liquid being used.

Annex: original figures

Figures produced by the experiment on 6 November 2026, unmodified. The curves of Figures 2 and 3 above use exactly these values.

